

2 Project 2: Convection Diffusion equation

2.1 Problem Formulation and Galerkin Approach

We consider the domain $\Omega = [0, 1]^2 = \{(x, y) \in \mathbb{R}^2 : 0 \leq x, y \leq 1\}$ and the stationary convection-diffusion equation with Dirichlet boundary conditions:

$$-D\Delta u + \beta \cdot \nabla u = 0 \quad \text{in } \Omega \subset \mathbb{R}^2 \quad (13)$$

$$u(x, y) = 1 \quad \text{on } \Gamma_1 = (\{0\} \times [0, 1]) \cup ([0, 0.3] \times \{0\}) \quad (14)$$

$$u(x, y) = 0 \quad \text{on } \Gamma_2 = (\{1\} \times [0, 1]) \cup ([0, 1] \times \{1\}) \cup ([0.3, 1] \times \{0\}) \quad (15)$$

where $D > 0$ is the diffusion coefficient (varying in the interval $[10^{-3}, 1]$) and $\beta = (1, 3)^T$ is a given constant advective velocity field.

Theoretically, when convection dominates diffusion ($D \ll |\beta|$), relying solely on the Galerkin $\mathcal{P}^1$ method (along with boundary lifting), as we did in Section 1, will induce violent oscillations in our numerical solutions. We can reduce these oscillations by refining the mesh. However, since we are limited to using one mesh only, we will add an artificial diffusion term.

Given a triangulation $\mathcal{T}_h(\Omega)$, we construct a numerical approximation using the Galerkin $\mathcal{P}^1$ method and boundary lifting. Let $u_h = u_h^0 + u_g$, where $u_g \in H^1(\Omega)$ is a boundary lifting function. We seek $u_h^0 \in \mathcal{V}_h$ such that:

$$a(u_h^0 + u_g, v_h) = 0 \implies a(u_h^0, v_h) = -a(u_g, v_h) \quad \forall v_h \in \mathcal{V}_h \quad (16)$$

The bilinear form $a(\cdot, \cdot)$ is defined as:

$$a(u, v) = \int_{\Omega} D \nabla u \cdot \nabla v + (\beta \cdot \nabla u) v \, dx \, dy \quad (17)$$

By restricting $a(\cdot, \cdot)$ to $H_0^1(\Omega) \times H_0^1(\Omega)$, continuity and coercivity are guaranteed, with coercivity bounded by $a(u, u) \geq \frac{D}{1+C_{\Omega}^2} \|u\|_{H^1}^2$ (from the fact that D and β are constants), ensuring the existence of a weak solution via the Lax-Milgram Theorem.

Note: we choose u_g to be 0 in every node except the ones of Dirichlet.

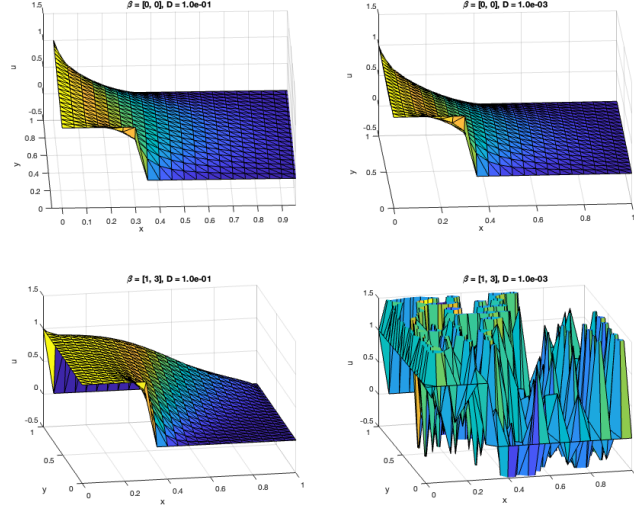
2.2 Implementation without stabilization

The standard Galerkin scheme leads to the linear system $(A + B)u = q$, where A is the stiffness matrix and B is the global transport (convection) matrix.

$B = (b_{ij})$ is obtained via the standard assembly method:

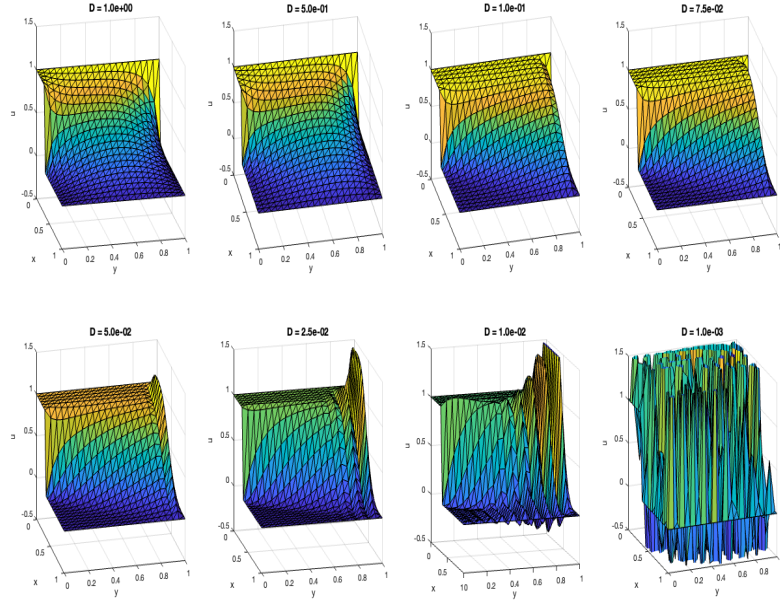
$$\begin{aligned} b_{ij} &= \sum_{T \in \mathcal{T}_h(\Omega)} b_{ij}^{(T)} = \sum_{T \in \mathcal{T}_h(\Omega)} \int_T (\beta \cdot \nabla \phi_j^{(T)}) \phi_i^{(T)} \, dx \, dy \\ &= \sum_{T \in \mathcal{T}_h(\Omega)} \left(\beta_1 b_j^{(T)} + \beta_2 c_j^{(T)} \right) \frac{\text{area}(T)}{3} \end{aligned}$$

We empirically notice that the presence of the new term ' β ' implies complications as the image below shows:



2.2.1 pseudo Peclet number

The convection term lead us to introduce a so called Peclet number (non dimensional) which has been proved to cause oscillations if bigger than 2 in dimension 1. In two or higher dimensions, particularly on unstructured meshes, establishing a strict universal definition of this number is challenging, making exact a priori stability estimates more complex. Therefore, we proceed by experimentally verifying the critical diffusion threshold for which the numerical problem remains stable. We introduce the notion of a local "pseudo Péclet number", defined as: $P_{ec} = \frac{|\beta|*h}{D}$ where $h \approx \max(2 * \sqrt{Area}) \approx 0.07$



Hence we notice that we lose stabilization for $D \leq 5e - 2$ and to say $P_{ec} \geq 4, 4$

2.3 Implementation of SUD

In order to stabilize our problem and smoothen the solution we add the Stabilization matrix to our equation, which consists in introducing an additional term, the numerical diffusion, along the stream- lines, so along the projection of the gradients in the direction of the vector field β . Our aim is to maintain consistency of the FEM scheme and to introduce the smallest necessary stabilizing terms. The problem becomes $(A+B+S)u = q$ where

$$s_{ij} = \sum_{T \in \mathcal{T}_h(\Omega)} s_{ij}^{(T)} = \delta * \sum_{T \in \mathcal{T}_h(\Omega)} \int_T (\beta \cdot \nabla \phi_j^{(T)}) (\beta \cdot \nabla \phi_i^{(T)}) dx dy \quad (18)$$

Where δ corresponds to $\tau \cdot \frac{P_{ec}}{|\beta|^2}$ and τ is searched experimentally. We notice that in the framework of our problem, i.e. $c=0$, $f=0$ and P_1 -Galerkin, implementing SD is equivalent to implementing SUPG.

2.4 Stabilization results

We now see how varying delta modifies the solutions of our problems for $D = [5e-2, 1e-2, 1e-3]$

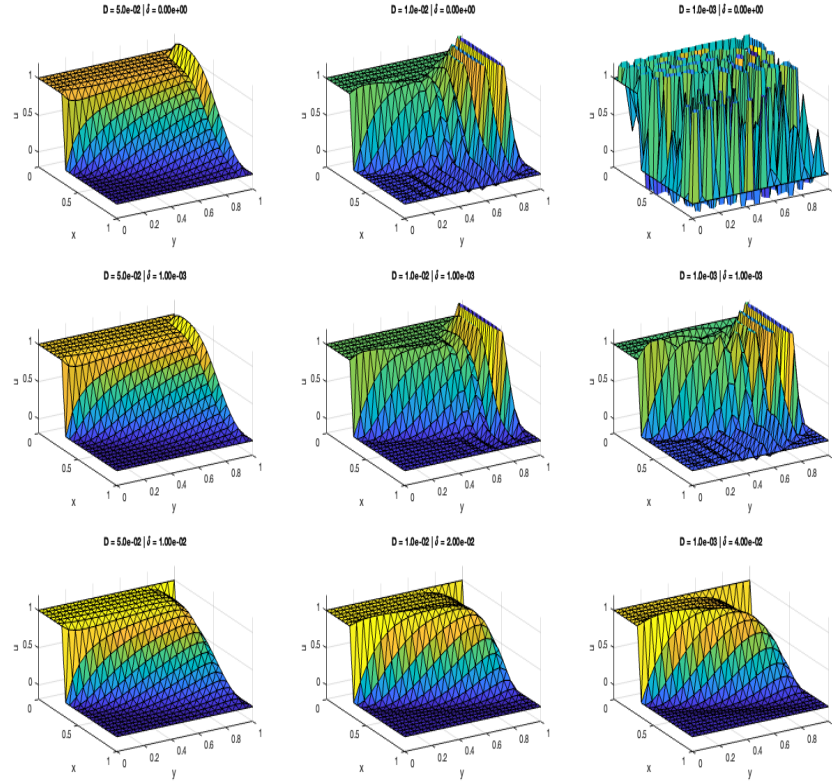


Figure 4: *The first row shows the solution without stabilization. The second the result for a small delta and the third the result for an nearly optimal one.*

We understand that the choice of delta has to be made with criterion. Two scenarios may happen:

- **Under-stabilization (δ too small)** Insufficient artificial diffusion is injected, allowing oscillations.
- **Over-stabilization (δ too big)** Excessive stabilization introduces an overabundance of artificial diffusion, fundamentally altering the physics of the original governing equation

Our ideal values stand in the middle.

3 project 3: Parabolic equation

3.1 Problem Formulation

We consider the domain

$$\Omega = [-1, 1]^2 = \{(x, y) \in \mathbb{R}^2 : -1 \leq x, y \leq 1\}$$

and the parabolic equation with Dirichlet boundary conditions:

$$\frac{\partial u}{\partial t} = \operatorname{div}(D \nabla u) + f(x, t) \quad \text{for } (x, t) \in \Omega \times [0, T], \quad (19)$$

$$u(x, 0) = u_0(x) \quad \text{for } x \in \Omega, \quad (20)$$

$$u(x, t) = 0 \quad \text{for } t > 0, x \in \partial\Omega. \quad (21)$$

We are given the manufactured solution $u(x, y, t) = e^{-t} \sin(\pi x) \cos(\frac{\pi}{2} y)$ and the terminal time $T = 1$. We Assume $D = 1$

We formulate the weak form of the parabolic problem by seeking $u \in L^2(0, T; H_0^1(\Omega))$ with time derivative $\dot{u} := \partial_t u \in L^2(0, T; H^{-1}(\Omega))$ such that:

$$(\dot{u}, v)_{L^2(t)} + a(u, v)(t) = (f, v)_{L^2(t)} \quad \forall v \in H_0^1(\Omega), \text{ for a.e. } t \in [0, T],$$

We then introduce a conforming $\mathcal{P}^1$ Galerkin approximation using the finite element space $\mathcal{V}_h = \operatorname{span}\{\phi_1, \dots, \phi_{N_h}\}$. We seek an approximate solution of the form:

$$u_h(x, y, t) = \sum_{j=1}^{N_h} C_j(t) \phi_j(x, y)$$

such that testing against any basis function ϕ_i yields:

$$\sum_{j=1}^{N_h} \dot{C}_j(t) (\phi_j, \phi_i)_{L^2} + \sum_{j=1}^{N_h} C_j(t) a(\phi_j, \phi_i) = (f(t), \phi_i)_{L^2} \quad \forall i = 1, \dots, N_h.$$

This can be formulated compactly in matrix form as a system of Ordinary Differential Equations (ODEs):

$$P \dot{u}(t) + A u(t) = b(t) \quad \text{for } t \in [0, T], \quad (22)$$

where $u(t) = (C_j(t))$ is the vector of nodal unknowns, and the mass matrix $P = (p_{ij})$, stiffness matrix $A = (a_{ij})$, and load vector $b(t) = (b_i(t))$ are defined by:

$$p_{ij} = (\phi_j, \phi_i)_{L^2}; \quad a_{ij} = a(\phi_j, \phi_i); \quad b_i(t) = (f(t), \phi_i)_{L^2}.$$

This represents a semi-discrete formulation, in contrast to the fully stationary discrete equations solved in the previous two projects. To obtain a fully discrete numerical approximation of $u_h(T)$, we apply the θ -method for time integration. Partitioning the interval $[0, T]$ uniformly into $\{t_0, t_1, \dots, t_N\}$ where $t_n = n\Delta t$, and setting $u^n \approx u(t_n)$, we rearrange the terms to solve for the unknown state at t_{n+1} .

$$(P + \theta \Delta t A) u^{n+1} = (P - \Delta t(1 - \theta)A) u^n + \theta b^{n+1} + (1 - \theta) b^n.$$